Embodied AI in Autonomous Driving Final Project

Retrieve-then-Rerank Inference for Vocabulary-Based End-to-End Driving

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Content

Introduction

1. E2E AD path panning
2. Problems of Vocabulary-based planning

End-to-End Autonomous Driving Path Planning

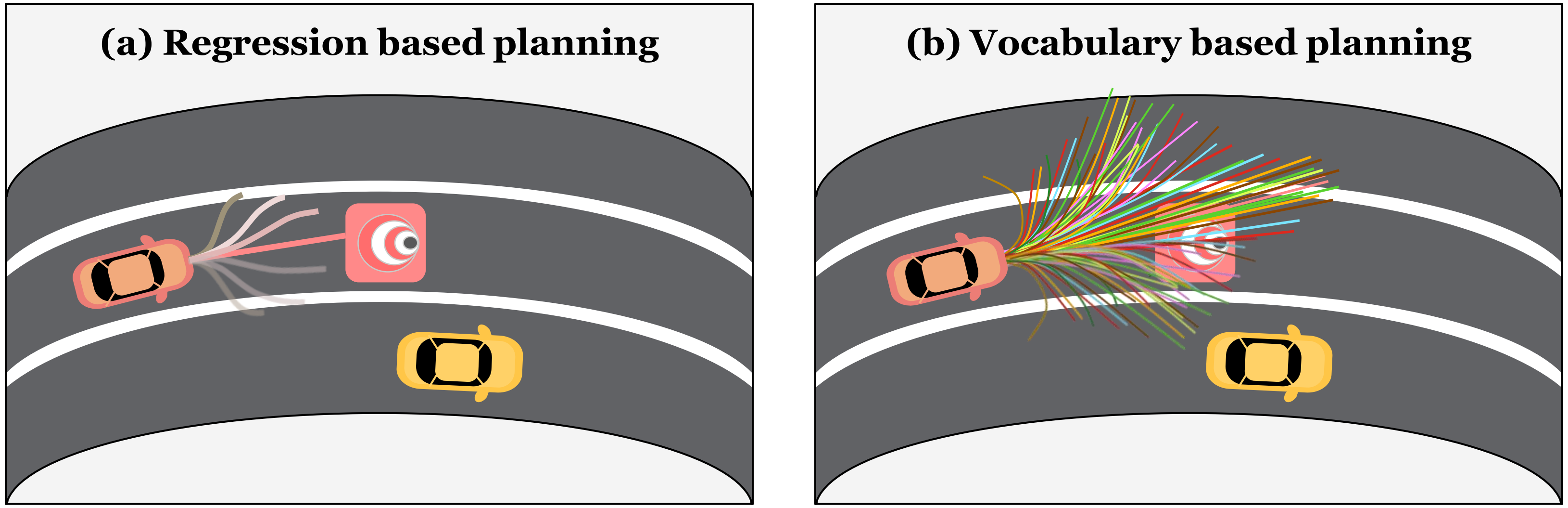
Approaches

▶ 1. Regression based planning

- Directly regress a single future trajectory from the current scene
  - ex. VAD, GenAD

▶ 2. Vocabulary based planning

- Select and score candidates from a fixed trajectory vocabulary
  - Hydra-MDP, VADv2



Path planning approaches in E2E AD

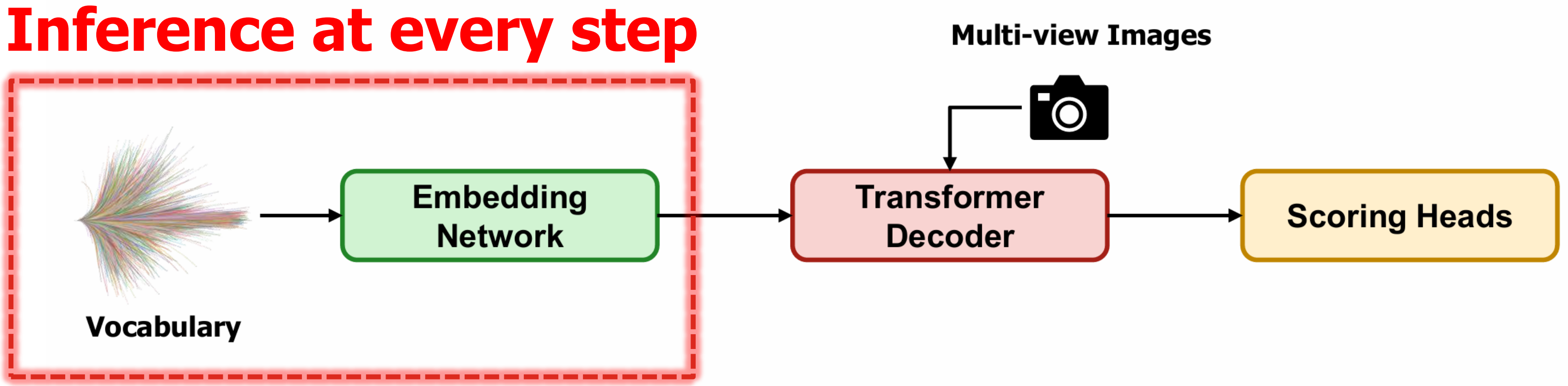
Vocabulary based Planning Overview

Methods

- Forward the fixed vocabulary embeddings at every step

Problem

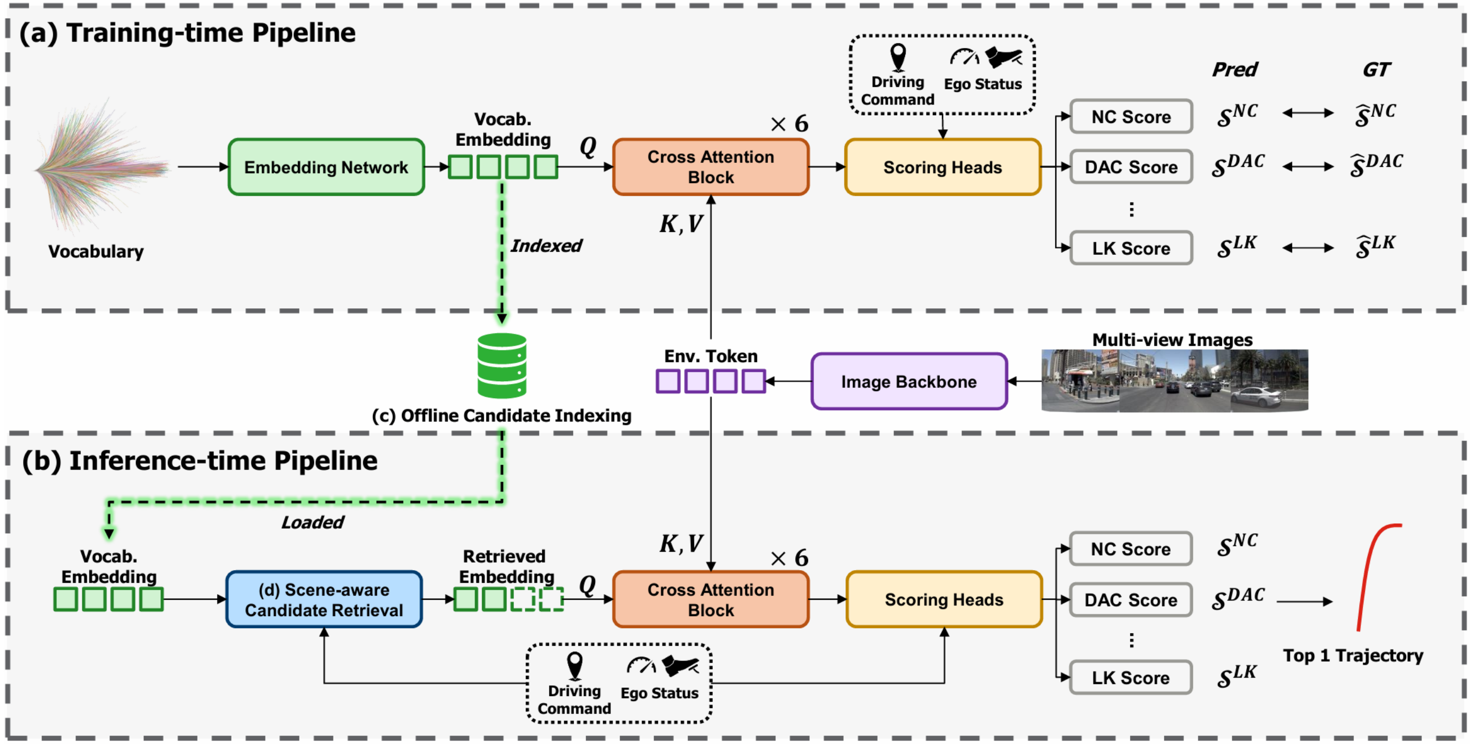
- As the reusable fixed candidates, the vocabulary embeddings, are inferred at every step, latency surges as size and complexity increase



Inference scheme of existing vocabulary-based E2E AD

Proposed Solution: Retrieve-then-Rerank

- **Phase1. Training**
  - Input: (Vocabulary), (Multi-view Images), (ego status), (driving command)
  - Output: (Cached vocabulary embedding), (Score)
- **Phase2. Inference**
  - Input : (Multi-view Images), (ego status), (driving command)
  - Output : (Score), (Top-1 Trajectory)



Methods

1. OCI (Offline Candidate Indexing)
2. SCR (Scene-aware Candidate Retrieval)

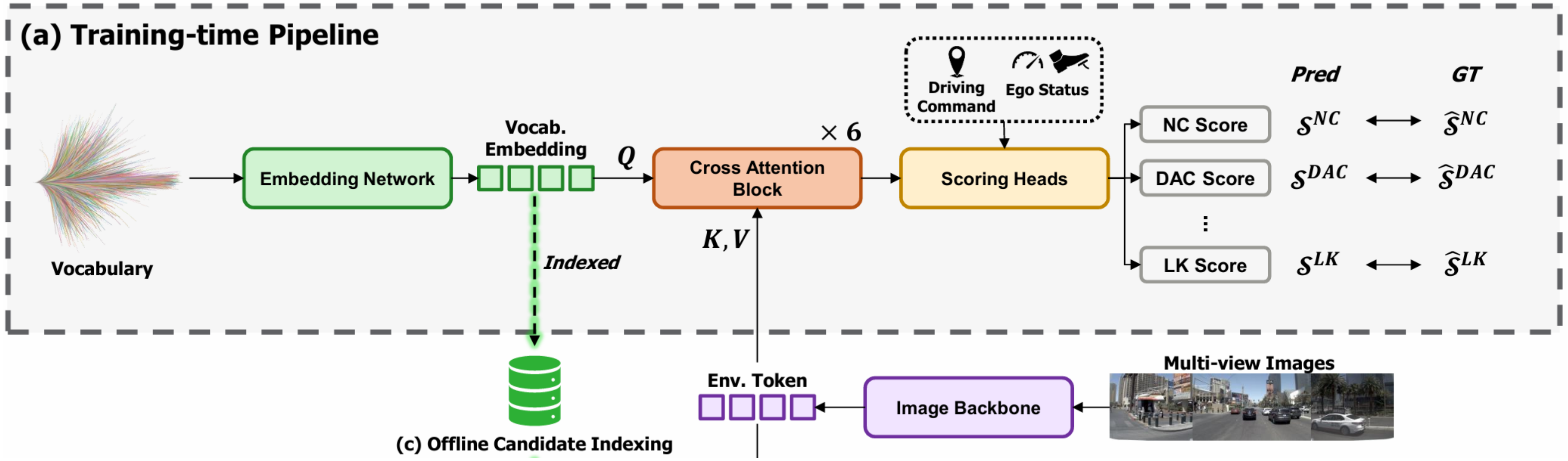
[Phase 1] Vocabulary based Planning Forward

Method

- No retrieval is used during training → forward the entire Vocabulary as-is to learn the Score

Score

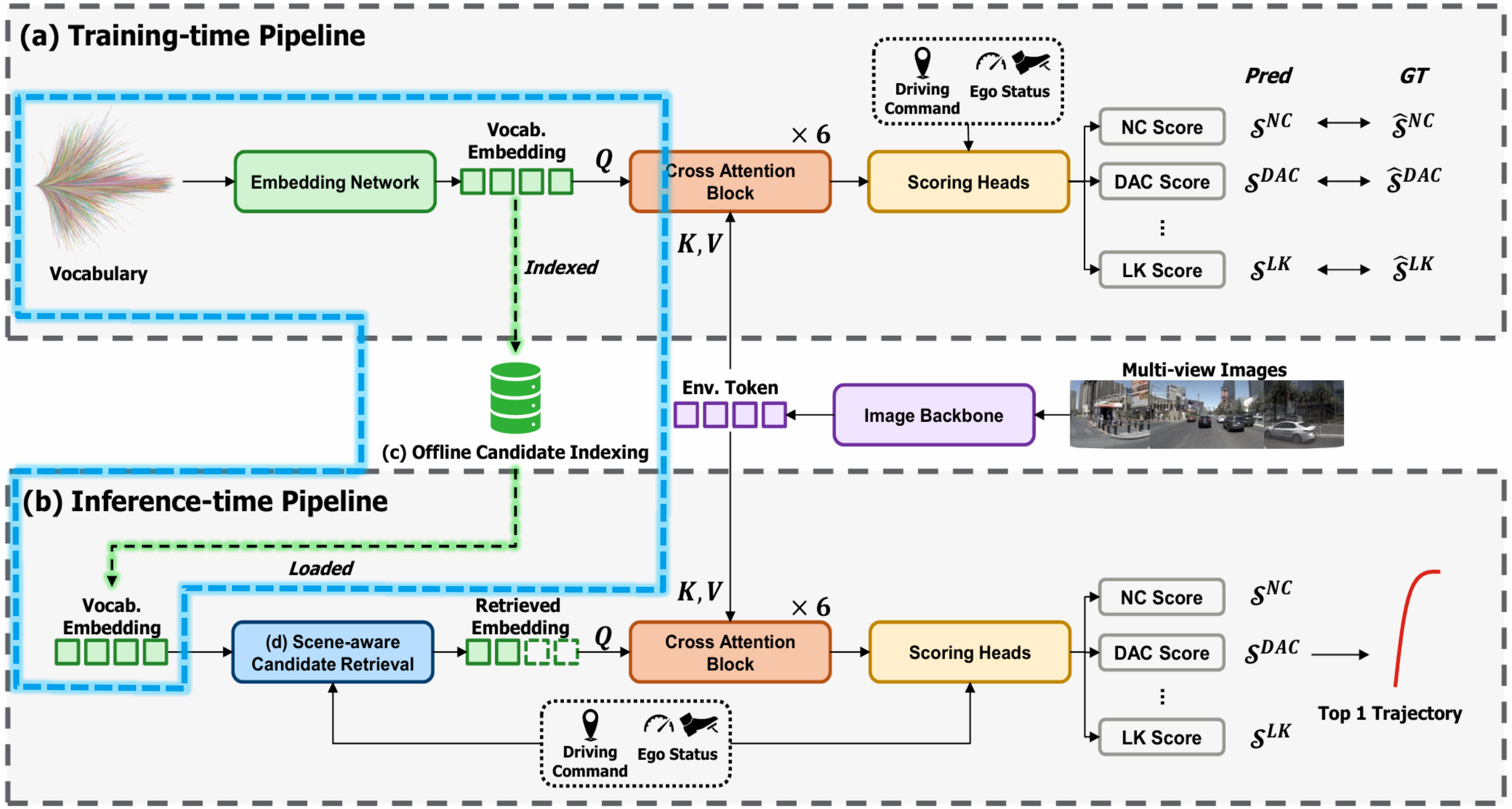
Score	Description	Score	Description
NC	No at fault collision	TTC	Time to collision within bound
DAC	Driving area compliance	LK	Lane keeping
TL	Traffic light compliance	HC	History comfort
EP	Ego progress	EC	Two frame and extended comfort



[Phase 2] OCI - Offline Candidate Indexing

Method

- **Index** the scene-invariant candidate embeddings **only once offline** and load them at inference **without recomputation**



OCI – Offline Candidate Tracking

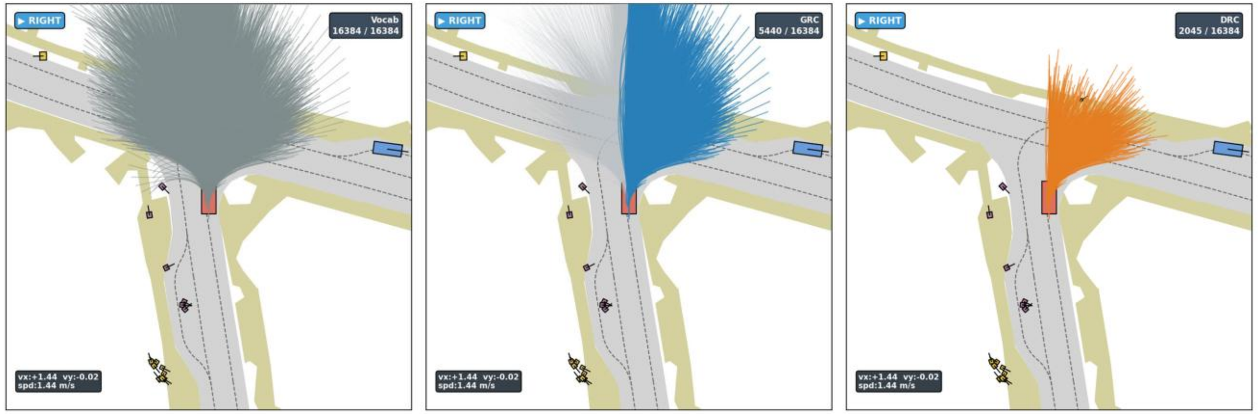
[Phase 2] SCR – Scene-aware Candidate Retrieval

Method

- A method to narrow the indexed Vocabulary in two stages before evaluation
- (Vocabulary) → [GRC] → [DRC] → [Candidates]

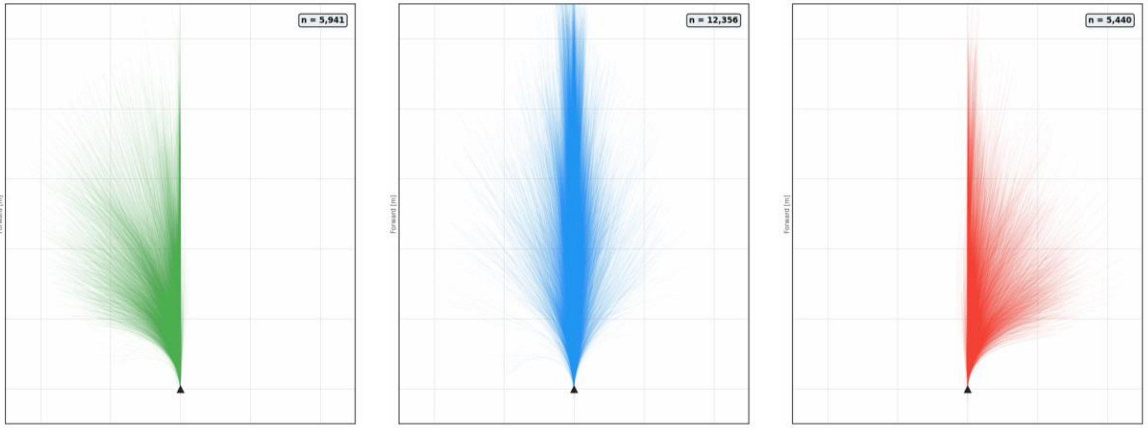
Score

- GRC – Global Route Compliance
  - Driving command (left turn / straight / right turn) → precomputed cluster lookup (cost ≈ 0)
- DRC – Dynamic Reachability Compliance
  - Dynamic reachability judgment based on ego velocity



(a) Full Vocabulary (b) After GRC Retrieval (c) After DRC Retrieval

SCR — Total 16,384 → GRC (right 5,440) → DRC 2,045



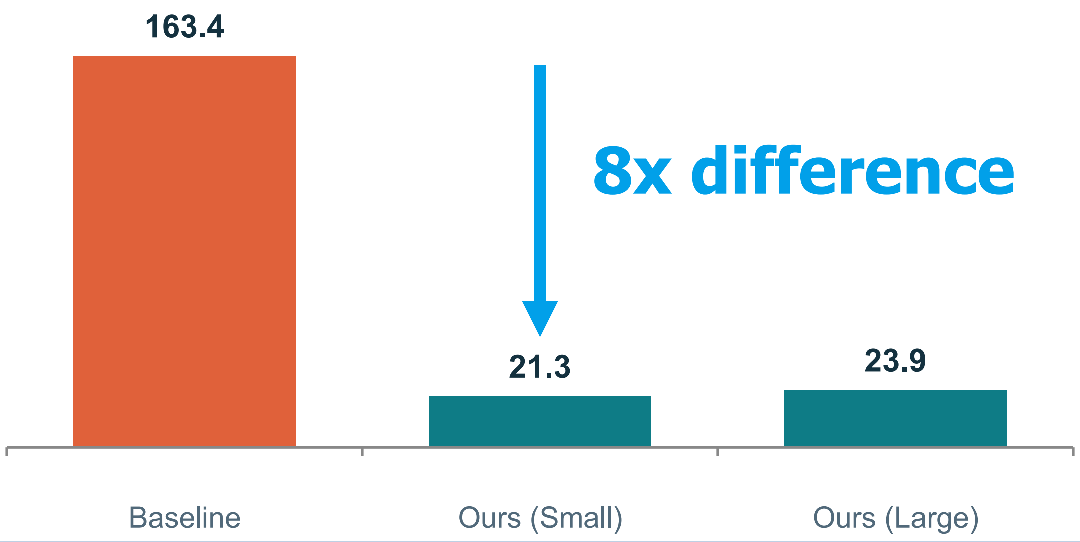
(a) Left cluster. (b) Straight cluster. (c) Right cluster.

clusters by left/straight/right command — GRC

Experiments

Experiments (1/4)

- **Training and inference environment**
  - 4x RTX 4090
  - batch size: 1
- **Dataset**
  - NAVSIM(navtrain/ navtest) + navhard
- **Metric**
  - EPDMS
    - NAVSIM Benchmark comprehensive evaluation metric



Latency 비교

Method	Vocab. Size	Backbone	Latency (ms)↓	NAVSIM v2 navhard EPDMS-1↑	NAVSIM v2 navhard EPDMS-2↑	NAVSIM v2 navhard EPDMS↑	NAVSIM v2 navtest EPDMS↑
Baseline	16,384	V2-99	163.4	74.1	48.9	37.6	86.9
Ours (Small)	8,192	V2-99	**21.3**	74.3	49.0	38.0	85.3
Ours (Large)	16,384	V2-99	23.9	75.2	50.1	40.2	86.1

Quantitative evaluation results – NAVSIM v2 navhard / navtest

Experiments (2/4)

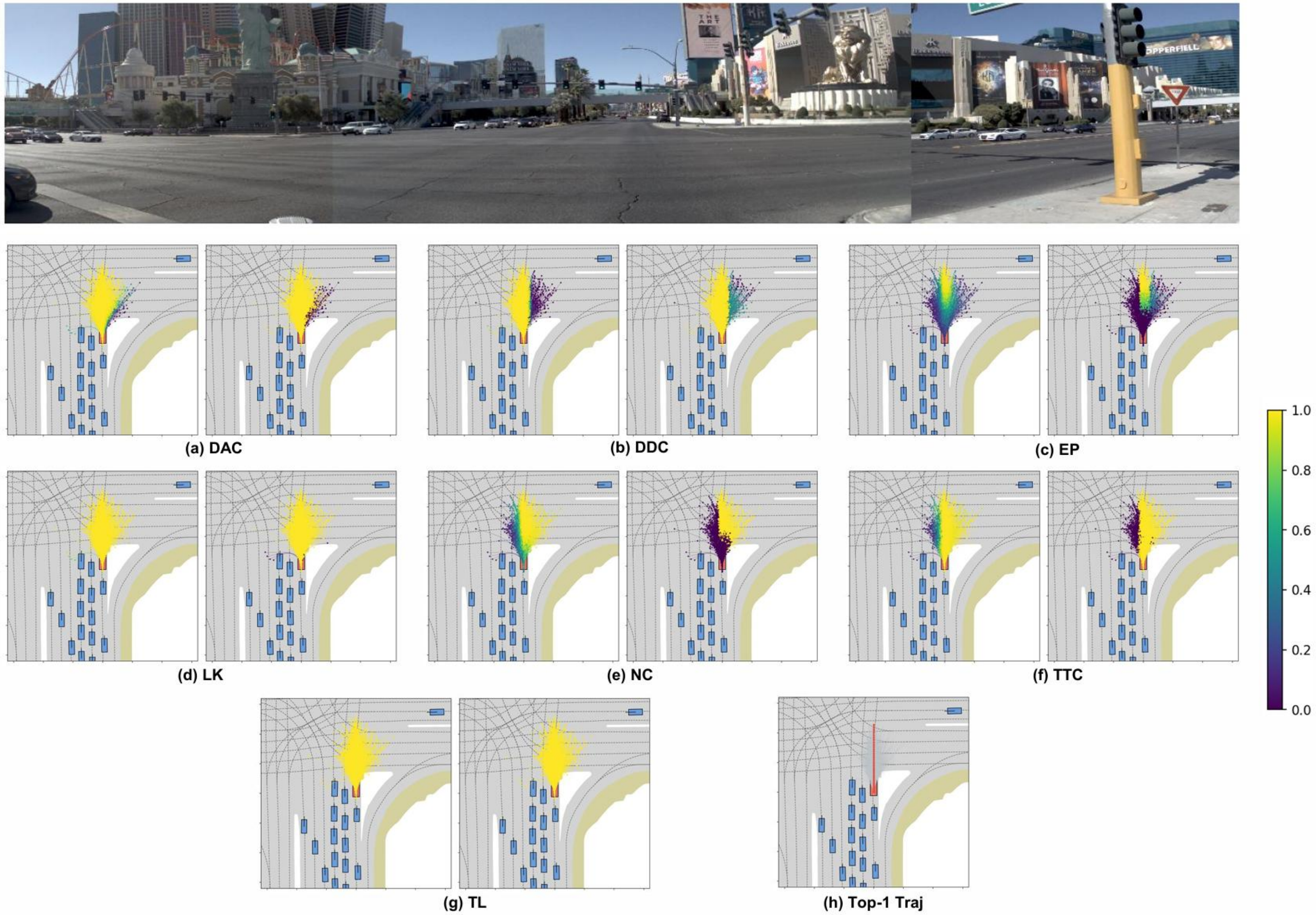
NAVSIM v2 – navtest evaluation results

- Slight performance drop versus baseline, **but 8× faster in terms of latency**

Method	Backbone	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑	EPDMS_{HF} ↑
Ego Status MLP	ResNet-34	93.1	77.9	92.7	99.6	86.0	91.5	89.4	98.3	85.4	64.0	–
TransFuser [4]	ResNet-34	96.9	89.9	97.8	99.7	87.1	95.4	92.7	98.3	87.2	76.7	–
DriveSuprim [27]	ResNet-34	97.5	96.5	99.4	99.6	88.4	96.6	95.5	98.3	77.0	83.1	–
DiffusionDriveV2 [29]	ResNet-34	97.7	96.6	99.2	99.8	88.3	97.2	96.0	97.8	91.0	85.5	–
ARTEMIS [7]	ResNet-34	98.3	95.1	98.6	99.8	81.5	97.4	96.5	98.3	98.3	83.1	–
Hydra-MDP++ [14]	V2-99	98.4	98.0	99.4	99.8	87.5	97.7	95.3	98.3	77.4	85.1	–
Baseline	V2-99	98.5	98.7	99.6	99.9	88.4	98.2	97.0	98.3	79.4	86.9	90.5
Ours (Small)	V2-99	98.1	98.0	99.4	99.9	87.8	97.5	96.3	98.2	77.9	85.3	88.7
Ours (Large)	V2-99	98.5	98.0	99.6	99.8	89.1	98.1	96.4	98.3	77.5	86.1	89.6

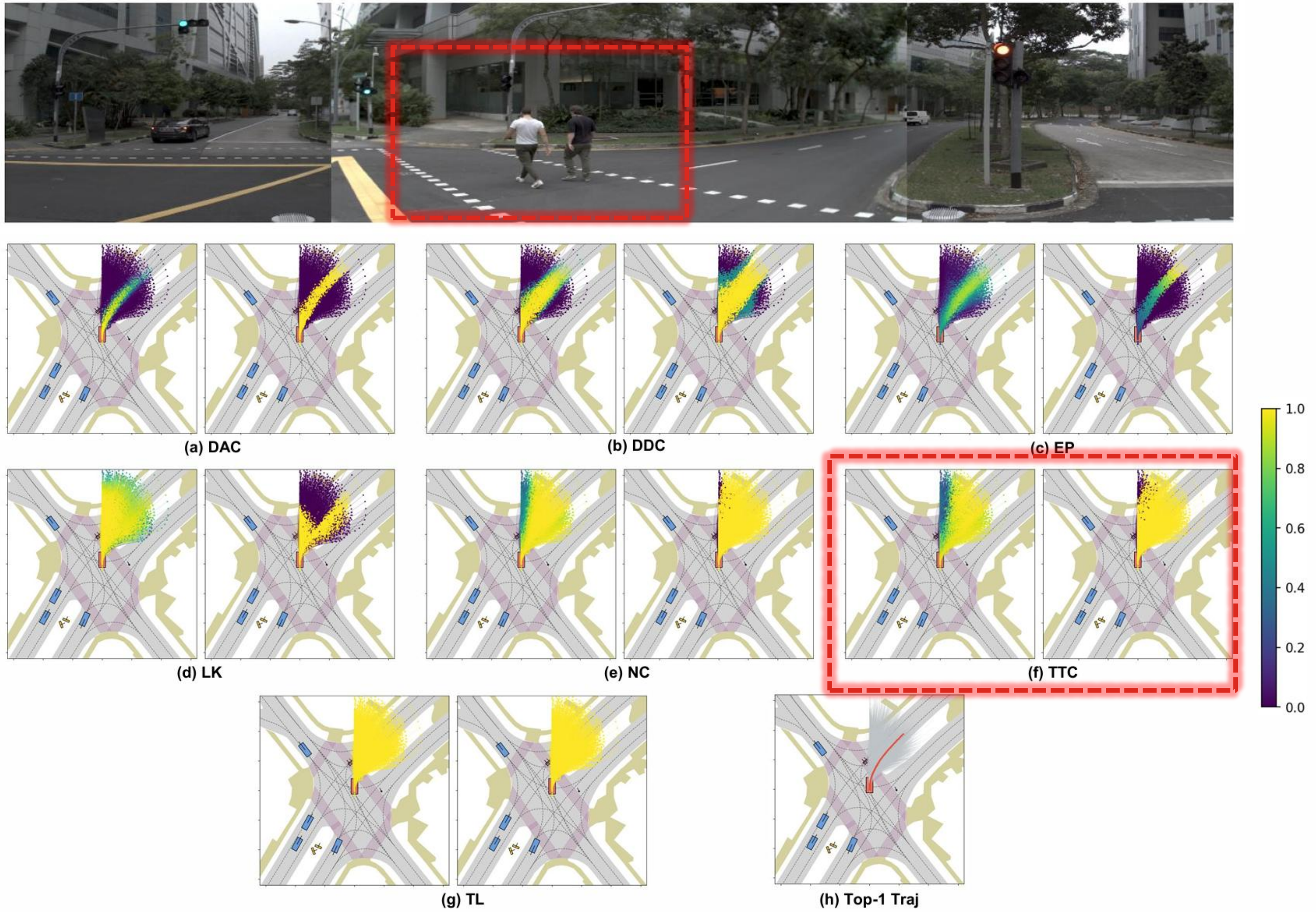
Quantitative evaluation results– NAVSIMv2 navtest

Experiments (3/4) – Straight Case



Qualitative Result – Straight Case

Experiments (4/4) – Turning Case



Qualitative Result – Turning Case

Conclusion

Conclusion

- **Retrieve-then-rerank framework**
  - Propose a method to quickly retrieve and rerank the existing fixed Vocabulary using OCI+SCR

- **Efficiency–quality trade-off improvement**
  - 163.4ms → 23.9ms (≈6.8x), navtest EPDMS 86.1 maintained, navhard 37.6 → 40.2 improved)

THANK YOU
FOR YOUR ATTENTION

Appendix A.

Official EPDMS Terms

EPDMS combines two groups of sub-scores. The first group consists of multiplicative compliance terms:

$$\mathcal{M}_{\text{pen}} = \{\text{NC}, \text{DAC}, \text{DDC}, \text{TL}\}, \tag{19}$$

where NC, DAC, DDC, and TL denote No at-fault Collision, Drivable Area Compliance, Driving Direction Compliance, and Traffic Light Compliance, respectively. These terms act as penalties because a severe violation directly reduces the final score.

The second group consists of weighted-average terms:

$$\mathcal{M}_{\text{avg}} = \{\text{EP}, \text{TTC}, \text{LK}, \text{HC}, \text{EC}\}, \tag{20}$$

where EP, TTC, LK, HC, and EC denote Ego Progress, Time-to-Collision, Lane Keeping, History Comfort, and Extended Comfort, respectively. These terms measure progress, safety margin, lane adherence, and comfort-related behavior after the selected trajectory is evaluated by the NAVSIM v2 scoring pipeline.

Let $s_m(\tau^*) \in [0, 1]$ be the official sub-score of the selected trajectory τ^* for metric term m, and let w_m be the benchmark-provided weight for each weighted-average term. The final EPDMS is computed as

$$\text{EPDMS}(\tau^*) = \left(\prod_{m \in \mathcal{M}_{\text{pen}}} s_m(\tau^*) \right) \cdot \frac{\sum_{m \in \mathcal{M}_{\text{avg}}} w_m s_m(\tau^*)}{\sum_{m \in \mathcal{M}_{\text{avg}}} w_m}. \tag{21}$$

In the official NAVSIM v2 setting, the weighted-average parameters are $w_{\text{EP}} = 5$, $w_{\text{TTC}} = 5$, $w_{\text{LK}} = 2$, $w_{\text{HC}} = 2$, and $w_{\text{EC}} = 2$.

All reported EPDMS values are computed from the selected final trajectory using the official NAVSIM evaluation code. Therefore, EPDMS should be interpreted as the external benchmark score, not as the learned score used internally to rank candidates during inference.

Appendix B.

Inference time score aggregation (selecting top 1 trajectory)

During inference, candidates are ranked by the same log-space aggregation rule used by the baseline selection-based planner:

$$
\begin{aligned}
f_\theta(\tau_i, \mathcal{S}) = {} & \lambda_{\mathrm{IMI}} \log p_i^{\mathrm{IMI}} + \lambda_{\mathrm{TL}} \log \sigma(u_i^{\mathrm{TL}}) + \lambda_{\mathrm{NC}} \log \sigma(u_i^{\mathrm{NC}}) + \lambda_{\mathrm{DAC}} \log \sigma(u_i^{\mathrm{DAC}}) \\
& + \lambda_{\mathrm{DDC}} \log \sigma(u_i^{\mathrm{DDC}}) + \lambda_{\mathrm{avg}} \log \left(\omega_{\mathrm{TTC}} \sigma(u_i^{\mathrm{TTC}}) + \omega_{\mathrm{EP}} \sigma(u_i^{\mathrm{EP}}) + \omega_{\mathrm{LK}} \sigma(u_i^{\mathrm{LK}})\right),
\end{aligned} \tag{22}
$$

where $\sigma(\cdot)$ denotes the sigmoid function. In our implementation, the aggregation parameters are $\lambda_{\mathrm{IMI}} = 0.03$, $\lambda_{\mathrm{TL}} = 0.10$, $\lambda_{\mathrm{NC}} = 0.10$, $\lambda_{\mathrm{DAC}} = 0.90$, $\lambda_{\mathrm{DDC}} = 0.20$, $\lambda_{\mathrm{avg}} = 6.00$, $\omega_{\mathrm{TTC}} = 7.00$, $\omega_{\mathrm{EP}} = 7.00$, and $\omega_{\mathrm{LK}} = 3.00$.

The final trajectory is selected as

$$
\tau^* = \arg\max_{\tau_i \in \mathcal{C}} f_\theta(\tau_i, \mathcal{S}). \tag{23}
$$

Appendix C.

Compute resources

Experiment	GPU setting	Batch / scope	Wall-clock time
Large-vocabulary training checkpoint	4× RTX 4090 24GB	batch 8, 20 epochs	∼20 h
Small-vocabulary training checkpoint	4× RTX 4090 24GB	batch 8, 20 epochs	∼20 h
Full-vocabulary baseline evaluation	1× RTX 4090 24GB	batch 1, navtest/navhard	<1 h per split
DRSI evaluation, Small/Large	1× RTX 4090 24GB	batch 1, navtest/navhard	<1 h per split
Retrieval and latency ablations	1× RTX 4090 24GB	batch 1, inference-only	<1 h per setting
Trajectory-geometry diagnostic	1× RTX 4090 24GB + CPU	sampled trajectory pairs/scenes	∼2 h